

DFT-Aware Verification for Scan-Based ICs

Abstract. Design for Testability (DFT) is widely adopted in modern integrated circuits to improve manufacturing test quality and fault observability. However, the introduction of scan chains, test modes, clock control logic, and test access mechanisms also increases verification complexity. This paper presents a DFT-aware verification methodology for scan-based digital integrated circuits. The proposed methodology integrates functional verification and test-mode verification throughout the design process to ensure the correctness of both normal operation and test-related logic. The verification flow focuses on several key DFT components, including scan enable control, scan chain shifting, capture operations, test clock generation, reset behavior, and test mode transitions. Assertion-based verification and constrained-random simulation are employed to detect potential design errors, such as incorrect scan data propagation, clock conflicts, illegal mode switching, and unstable reset responses. In addition, functional coverage and code coverage are used to evaluate the completeness of the verification environment. The results indicate that the proposed methodology can effectively improve the observability of DFT-related defects and reduce the risk of test-mode failures before tape-out. This work provides a practical verification framework for improving the reliability and testability of digital chip designs.